

Unit-4

Computer Programming Fundamentals

Computer Network

- **Computer Network** is a group of computers connected with each other through wires, optical fibres or optical links so that various devices can interact with each other through a network.
- The aim of the computer network is the sharing of resources among various devices.
- In the case of computer network technology, there are several types of networks that vary from simple to complex level.

Uses Of Computer Network

1.Resource sharing: Resource sharing is the sharing of resources such as programs, printers, and data among the users on the network without the requirement of the physical location of the resource and user.

2.Server-Client model: Computer networking is used in the server-client model. A server is a central computer used to store the information and maintained by the system administrator. Clients are the machines used to access the information stored in the server remotely.

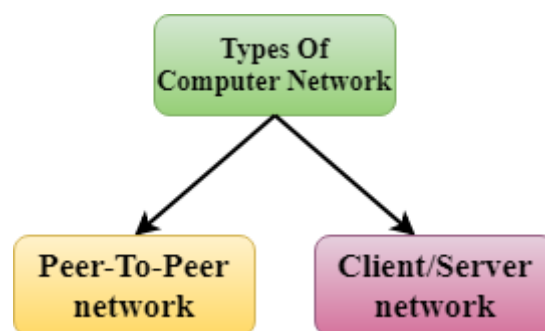
3.Communication medium: Computer network behaves as a communication medium among the users. For example, a company contains more than one computer has an email system which the employees use for daily communication.

4.E-commerce: Computer network is also important in businesses. We can do business over internet. For example, amazon is doing their business over internet.

Computer Network Architecture

Computer Network Architecture is defined as the physical and logical design of the software, hardware, protocols, and media of the transmission of data. Simply we can say that how computers are organized and how tasks are allocated to the computer.

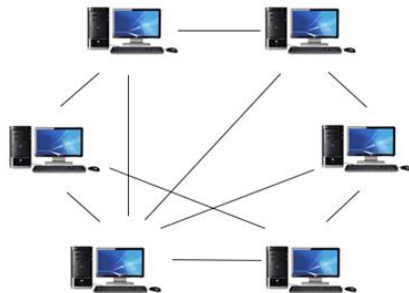
The two types of network architectures are used:



- Peer-To-Peer network
- Client/Server network

Peer-To-Peer network

- Peer-To-Peer network is a network in which all the computers are linked together with equal privilege and responsibilities for processing the data.
- Peer-To-Peer network is useful for small environments, usually up to 10 computers.
- Peer-To-Peer network has no dedicated server.
- Special permissions are assigned to each computer for sharing resources.



Advantages Of Peer-To-Peer Network:

- Its less costly as it doesn't contain dedicated server.
- If one computer stops working, other computers will not stop working.
- Its easy to set up and maintain as each computer manages itself.

Disadvantages Of Peer-To-Peer Network:

- It cannot back up data as data is different in different locations.
- It has a security issue as device is managed itself.

Client/Server Network

- Client/Server network is a network model designed for end users called clients, to access the resources such as songs, video, etc. from a central computer known as Server.
- The central controller is known as **server** while all other computers in the network are called **clients**.
- A server performs all the major operations such as security and network management.
- A server is responsible for managing all the resources such as files, directories, printer, etc.
- All clients communicate with each other through a server. For example, if client1 wants to send some data to client 2, then it first sends request to the server for permission. Server sends response to client1 to initiate its communication with client 2.



Advantages Of Client/Server network:

- A Client/Server network contains the centralized system. Therefore we can back up the data easily.

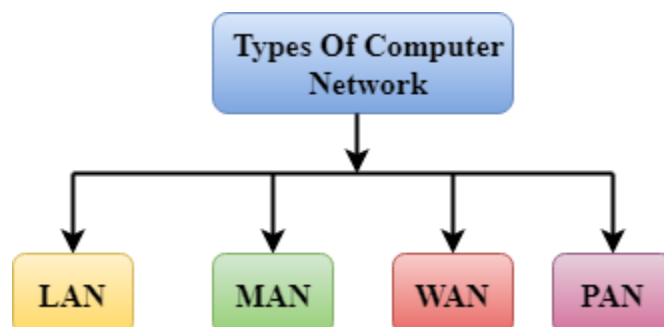
- A Client/Server network has a dedicated server that improves overall performance of system.
- Security is better in Client/Server network as a single server administers the shared resources.
- It also increases the speed of sharing resources.

Disadvantages Of Client/Server network:

- Client/Server network is expensive as it requires the server with large memory.
- It requires a dedicated network administrator to manage all the resources.

Computer Network Types

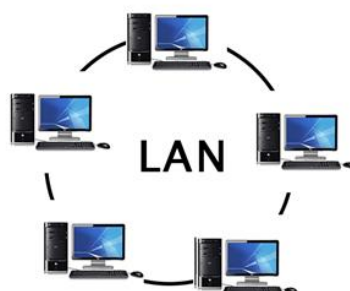
A computer network is a group of computers linked to each other that enable computer to communicate with another computer and share their resources, data, and applications. A computer network can be categorized by their size. A **computer network** is mainly of **four types**:



- LAN(Local Area Network)
- PAN(Personal Area Network)
- MAN(Metropolitan Area Network)
- WAN(Wide Area Network)

1.LAN(Local Area Network)

- Local Area Network is a group of computers connected to each other in a small area such as building, office.
- LAN is used for connecting two or more personal computers through a communication medium such as twisted pair, coaxial cable, etc.
- It is less costly as it is built with inexpensive hardware such as hubs, network adapters, and ethernet cables.
- The data is transferred at an extremely faster rate in Local Area Network.
- Local Area Network provides higher security.



2.PAN(Personal Area Network)

- Personal Area Network is a network arranged within an individual person, typically within a range of 10 meters.
- Personal Area Network is used for connecting the computer devices of personal use is known as Personal Area Network.
- Personal Area Network covers an area of **30 feet**.
- Personal computer devices that are used to develop the personal area network are the laptop, mobile phones, media player and play stations.

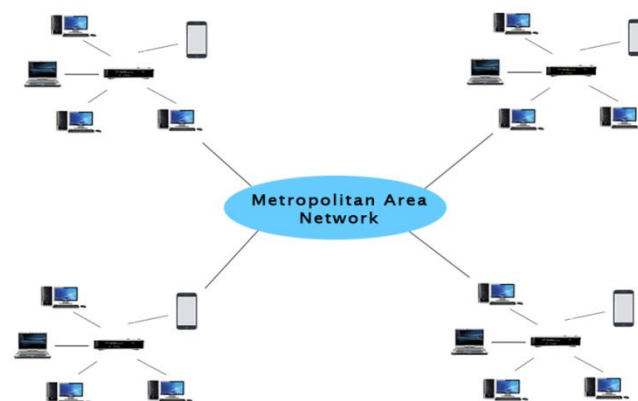


Wireless Personal Area Network: Wireless Personal Area Network is developed by using wireless technologies such as WiFi, Bluetooth.

Wired Personal Area Network: Wired Personal Area Network is created by using the USB.

3.MAN(Metropolitan Area Network)

- A metropolitan area network is a network that covers a larger geographic area by interconnecting different LAN to form a larger network.
- Government agencies use MAN to connect to the citizens and private industries.
- In MAN, various LANs are connected to each other through a telephone exchange line.
- The most widely used protocols in MAN are RS-232, Frame Relay, ATM, ISDN, OC-3, ADSL, etc.
- It has higher range than Local Area Network(LAN).



Uses Of Metropolitan Area Network:

- It is used in communication between banks in city.
- It can be used in an Airline Reservation.
- It can be used in a college within a city.
- It can be used for communication in the military.

4.WAN(Wide Area Network)

- A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- Wide Area Network is quite bigger network
- WAN is not limited to a single location, but it spans over large geographical area through a telephone line, fibre optic cable or satellite links.
- Internet is one of the biggest WAN in the world.
- A WAN is widely used in the field of Business, government, and education.

Examples Of Wide Area Network:

- **Mobile Broadband:** A 4G network is widely used across a region or country.
- **Last mile:** A telecom company is used to provide the internet services to the customers in hundreds of cities by connecting their home with fiber.

Advantages Of Wide Area Network:

- **Geographical area:** A Wide Area Network provides a large geographical area. Suppose if branch of our office is in different city then we can connect with them through WAN.

- **Centralized data:** In case of WAN network, data is centralized.
- **Get updated files:** Software companies work on live server. So, programmers get updated files.
- **Exchange messages:** In WAN network, messages are transmitted fast. The web application like Whatsapp allows to communicate with friends.
- **Sharing of software and resources:** In WAN network, we can share the software and other resources like a hard drive, RAM.
- **Global business:** We can do business globally.

Disadvantages of Wide Area Network:

- **Security issue:** A WAN network has less security as compared to LAN,MAN as all technologies are combined together that creates security problem.
- **Needs Firewall & antivirus software:** The data is transferred on internet which can be changed or hacked so firewall needs to be used.
- **High Setup cost:** Installation cost of WAN network is high as it involves purchasing of routers, switches.
- **Troubleshooting problems:** It covers a large area so fixing the problem is difficult.

Network Topology

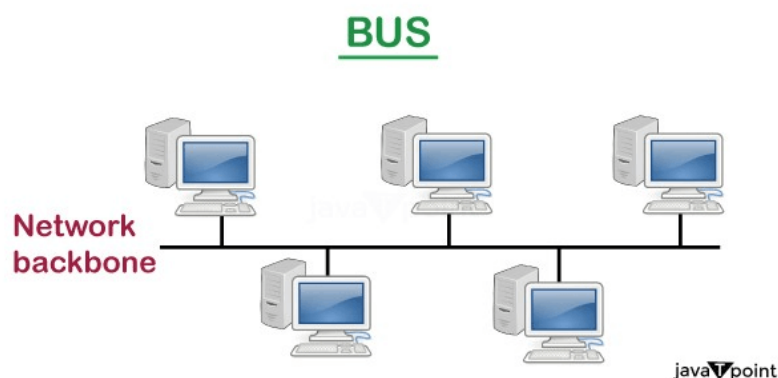
Topology defines the structure of the network of how all the components are interconnected to each other.

Types of Network Topology

Topology is geometric representation of all nodes in a network. There are six types of network topology

1. Bus Topology
2. Ring Topology
3. Tree Topology
4. Star Topology
5. Mesh Topology
6. Hybrid Topology

1) Bus Topology



- The bus topology is designed in such a way that all the stations are connected through a single cable known as a **backbone cable**.

- Each node is either connected to backbone cable through drop cable or directly connected to backbone cable.
- When a node wants to send a message over the network, it puts a message over network. All the stations available in the network will receive the message whether it has been addressed or not.
- The configuration of bus topology is simpler as compared to other topologies.
- Backbone cable is considered as a "**single lane**" through which message is broadcast to all stations.

Advantages of Bus topology:

- **Low-cost cable:** In bus topology, nodes are directly connected to cable without passing through a hub. Therefore, the initial cost of installation is low.
- **Moderate data speeds:** Coaxial or twisted pair cables are mainly used in bus-based networks.
- **Familiar technology:** Bus topology is a familiar technology as installation and troubleshooting techniques are well known, and hardware components are easily available.

- **Limited failure:** A failure in one node will not have any effect on other nodes.

Disadvantages of Bus topology:

- **Extensive cabling:** A bus topology is quite simpler, but still it requires a lot of cabling.
- **Difficult troubleshooting:** It requires specialized test equipment to determine the cable faults.
- **Signal interference:** If two nodes send the messages simultaneously, then the signals of both the nodes collide with each other.
- **Reconfiguration difficult:** Adding new devices to the network would slow down the network.
- **Attenuation:** Attenuation is a loss of signal leads to communication issues. Repeaters are used to regenerate the signal.

2) Ring Topology



- Ring topology is like a bus topology, but with connected ends.
- The node that receives message from the previous computer will retransmit to the next node.
- The data flows in one direction.
- The data flows in a single loop continuously known as an endless loop.
- It has no terminated ends, i.e., each node is connected to other node and having no termination point.
- The data in a ring topology flow in a clockwise direction.

Advantages of Ring topology:

- **Network Management:** Faulty devices can be removed from network without bringing network down.
- **Cost:** Twisted pair cabling is inexpensive and easily available. Therefore installation cost is very low.
- **Reliable:** It is a more reliable network because the communication system is not dependent on the single host computer.

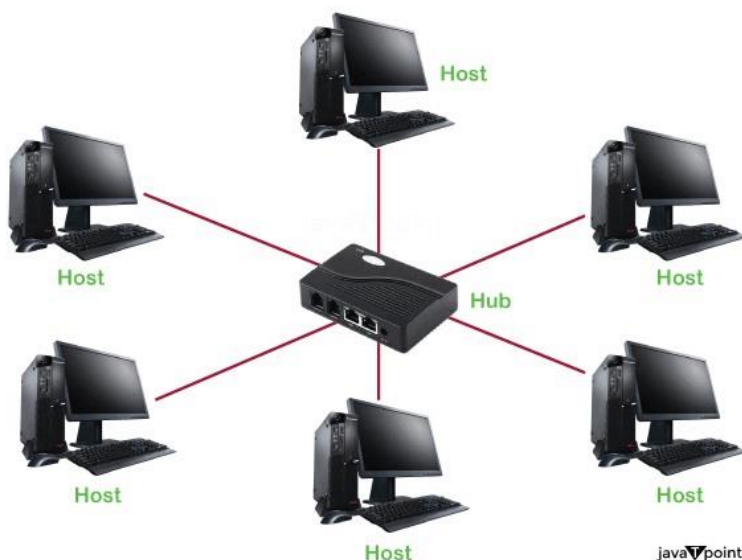
Disadvantages of Ring topology:

- **Difficult troubleshooting:** It requires specialized test equipment to determine the cable faults. If any fault occurs in the cable, then it would disrupt the communication for all the nodes.
- **Failure:** The breakdown in one station leads to the failure of the overall network.
- **Reconfiguration difficult:** Adding new devices to the network would slow down the network.
- **Delay:** Communication delay is directly proportional to the number of nodes. Adding new devices increases the communication delay.

3) Star Topology

Star Topology

- All hosts in Star topology are connected to a central device, known as hub device, using a point-to-point connection.
- Point-to-point connection between hosts and hub.



- Star topology is an arrangement of the network in which every node is connected to the central hub, switch or a central computer.
- Central computer is called **server**, and peripheral devices attached to server are known as **clients**.
- Coaxial cable or RJ-45 cables are used to connect the computers.
- Hubs or Switches are mainly used as connection devices in a **physical star topology**.
- Star topology is the most popular topology in network implementation.

Advantages of Star topology

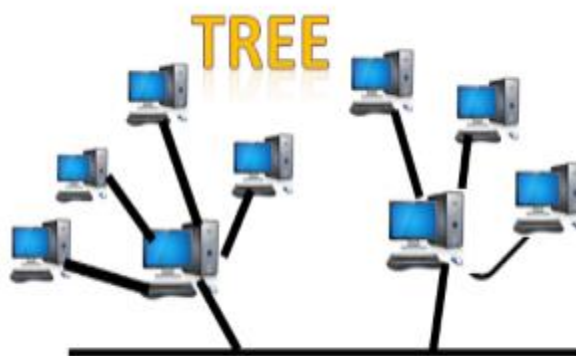
- **Efficient troubleshooting:** In star topology, all stations are connected to centralized network. Therefore network administrator has to go to the single station to troubleshoot the problem.
- **Network control:** Network control features can be easily implemented in star topology.
- **Limited failure:** As each station is connected to the central hub with its own cable, therefore failure in one cable will not affect the entire network.
- **Familiar technology:** Star topology is a familiar technology as its tools are cost-effective.

- **Easily expandable:** It is easily expandable as new stations can be added to the open ports on hub.
- **Cost effective:** Star topology networks are cost-effective as it uses inexpensive coaxial cable.
- **High data speeds:** It supports a bandwidth of approx 100Mbps. Ethernet 100BaseT is one of the most popular Star topology networks.

Disadvantages of Star topology

- **A Central point of failure:** If the central hub or switch goes down, then all the connected nodes will not be able to communicate with each other.
- **Cable:** Sometimes cable routing becomes difficult when a significant amount of routing is required.

4) Tree topology



- Tree topology combines the characteristics of bus topology and star topology.

- A tree topology is type of structure in which all the computers are connected in hierarchical fashion.
- The top-most node in tree topology is known as a **root node** and all other nodes child of root node.
- There is only one path exists between two nodes for data transmission. Thus, it forms a parent-child hierarchy.

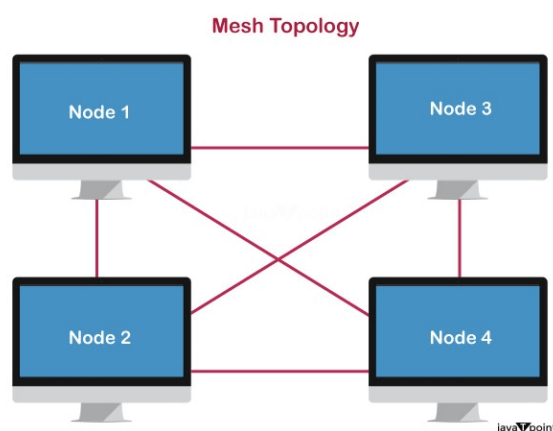
Advantages of Tree topology

- **Support for broadband transmission:** Tree topology is mainly used to provide broadband transmission, i.e., signals are sent over long distances without being attenuated.
- **Easily expandable:** We can add new device to the existing network. Therefore it is easily expandable.
- **Easily manageable:** In this, whole network is divided into segments known as star networks which can be easily managed and maintained.
- **Error detection:** Error detection and error correction are very easy in a tree topology.
- **Limited failure:** The breakdown in one station does not affect the entire network.
- **Point-to-point wiring:** It has point-to-point wiring for individual segments.

Disadvantages of Tree topology

- **Difficult troubleshooting:** If any fault occurs in the node, then it becomes difficult to find problem.
- **High cost:** Devices required for broadband transmission are very costly.
- **Failure:** A tree topology mainly relies on main bus cable and failure in main bus cable will damage the overall network.
- **Reconfiguration difficult:** If new devices are added, then it becomes difficult to reconfigure.

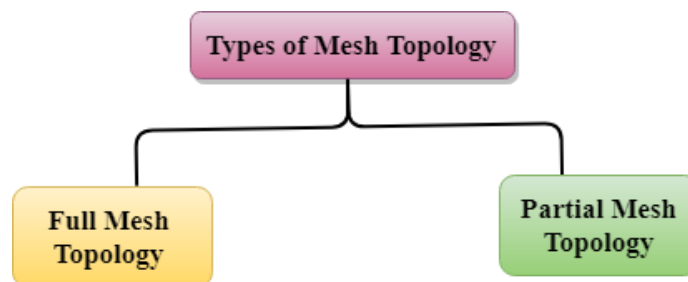
5) Mesh topology



- Mesh technology is arrangement of the network in which computers are interconnected with each other through various redundant connections.
- There are multiple paths from one computer to another computer.
- It does not contain switch, hub or any central computer.

- The Internet is an example of the mesh topology.
- Mesh topology is mainly used for WAN implementations.
- Mesh topology is mainly used for wireless networks.

Mesh topology is divided into two categories:



- **Full Mesh Topology:** In a full mesh topology, each computer is connected to all computers available in the network.
- **Partial Mesh Topology:** In a partial mesh topology, not all but certain computers are connected to those computers with which they communicate frequently.

Advantages of Mesh topology:

Reliable: The mesh topology networks are very reliable as if any link breakdown will not affect the communication between connected computers.

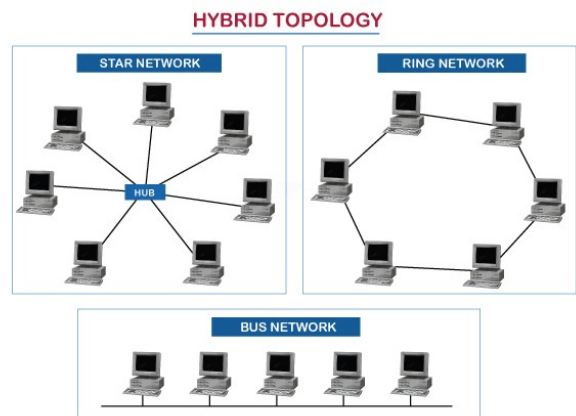
Fast Communication: Communication is very fast between the nodes.

Easier Reconfiguration: Adding new devices would not disrupt the communication between other devices.

Disadvantages of Mesh topology

- **Cost:** A mesh topology contains a large number of connected devices such as a router and more transmission media than other topologies.
- **Management:** Mesh topology networks are very large and very difficult to maintain and manage.
- **Efficiency:** In this topology, redundant connections are high that reduces efficiency of the network.

6) Hybrid Topology



- The combination of various different topologies is known as **Hybrid topology**.
- A Hybrid topology is a connection between different links and nodes to transfer the data.

- When two or more different topologies are combined together is termed as Hybrid topology. For example, if there exist a ring topology in one branch of ICICI bank and bus topology in another branch of ICICI bank, connecting these two topologies will result in Hybrid topology.

Advantages of Hybrid Topology

- **Reliable:** If fault occurs in any part of the network will not affect functioning of the rest of network.
- **Scalable:** Size of the network can be easily expanded by adding new devices without affecting the functionality of the existing network.
- **Flexible:** This topology is very flexible as it can be designed according to requirements of the organization.
- **Effective:** Hybrid topology is very effective as it can be designed such that strength of network is maximized and weakness of network is minimized.

Disadvantages of Hybrid topology

- **Complex design:** The major drawback of the Hybrid topology is design of Hybrid network. It is

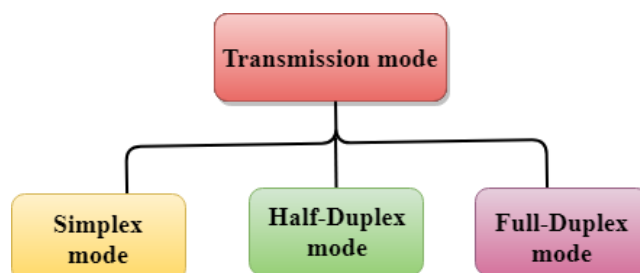
very difficult to design the architecture of the Hybrid network.

- **Costly Hub:** The Hubs used in the Hybrid topology are very expensive as these hubs are different from usual Hubs used in other topologies.
- **Costly infrastructure:** The infrastructure cost is very high as a hybrid network requires a lot of cabling, network devices, etc.

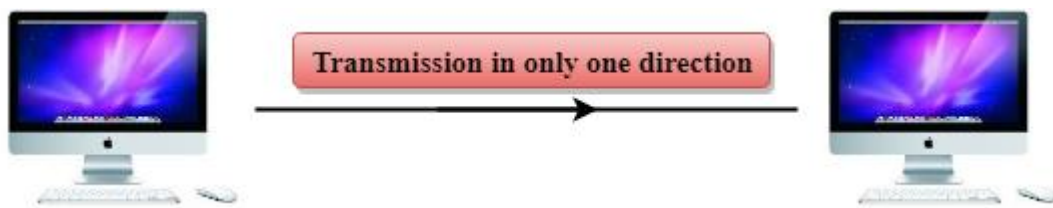
Transmission modes

- The way in which data is transmitted from one device to another device is known as **transmission mode**.
- The transmission mode is also known as the communication mode.
- Each communication channel has a direction associated with it, and transmission media provide the direction. Therefore, the transmission mode is also known as a directional mode.

The Transmission mode is divided into three categories:



1.Simplex mode



- In Simplex mode, the communication is unidirectional, i.e. the data flow in one direction.
- A device can only send the data but cannot receive it or it can receive data but cannot send the data.
- This transmission mode is not very popular as mainly communications require two-way exchange of data. The simplex mode is used in the business field as in sales that do not require any corresponding reply.
- The **radio station** is a simplex channel as it transmits the signal to the listeners but never allows them to transmit back.
- **Keyboard and Monitor** are examples of simplex mode.

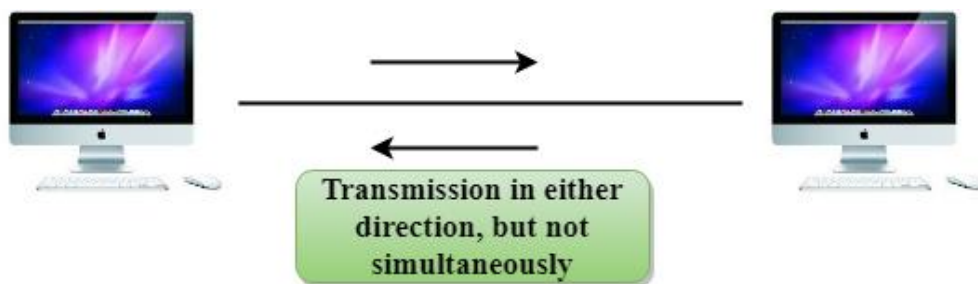
Advantage of Simplex mode:

- In simplex mode, the station can utilize the entire bandwidth of the communication channel, so that more data can be transmitted at a time.

Disadvantage of Simplex mode:

- Communication is unidirectional, so it has no inter-communication between devices.

2. Half-Duplex mode



- In Half-duplex channel, direction can be reversed, i.e. station can transmit and also receive the data.
- Messages flow in both the directions, but not at the same time.
- The entire bandwidth of the communication channel is utilized in one direction at a time.
- In half-duplex mode, it is possible to perform error detection, and if any error occurs, then the receiver requests sender to retransmit the data.
- A **Walkie-talkie** is an example of the Half-duplex mode. In Walkie-talkie, one party speaks, and another party listens. After a pause, other speaks and first party listens.

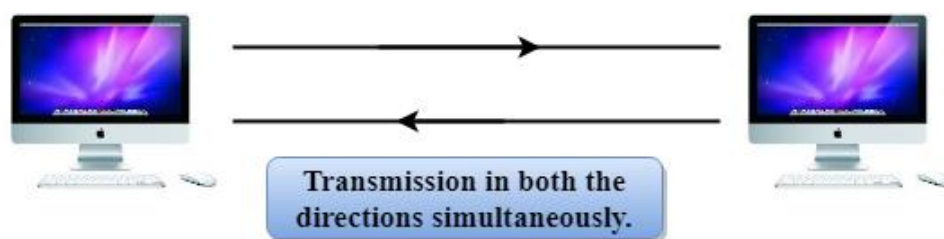
Advantage of Half-duplex mode:

- In half-duplex mode, both devices can send and receive data and can utilize entire bandwidth of communication channel during data transmission.

Disadvantage of Half-Duplex mode:

- In half-duplex mode, when one device is sending the data, then another has to wait, this causes the delay in sending the data at the right time.

3.Full-duplex mode



- In Full duplex mode, communication is bi-directional, i.e data flow in both the directions.
- Both stations can send receive message simultaneously.
- Full-duplex mode has two channels. One channel has traffic moving in one direction, and another channel has traffic moving in opposite direction.
- The Full-duplex mode is the fastest mode of communication between devices.
- The most common example of full-duplex mode is a **telephone network**. When two people are

communicating, both can talk and listen at the same time.

Advantage of Full-duplex mode:

- Both the stations can send and receive the data at the same time.

Disadvantage of Full-duplex mode:

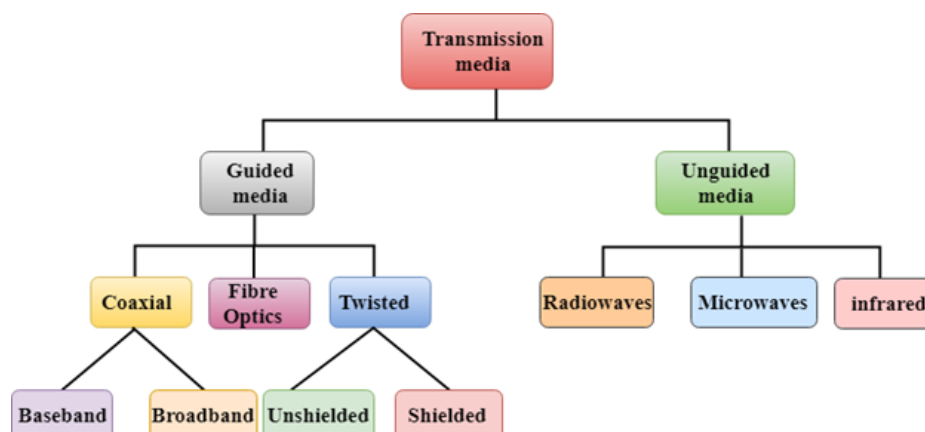
- If there is no dedicated path exists between the devices, then the capacity of the communication channel is divided into two parts.

Basis for comparison	Simplex mode	Half-duplex mode	Full-duplex mode
Direction of communication	In simplex mode, the communication is unidirectional.	In half-duplex mode, the communication is bidirectional, but one at a time.	In full-duplex mode, the communication is bidirectional.
Send/Receive	A device can only send the data but cannot receive it or it can only receive the data but cannot send it.	Both the devices can send and receive the data, but one at a time.	Both the devices can send and receive the data simultaneously.
Performance	The performance of half-duplex mode is better than the simplex mode.	The performance of full-duplex mode is better than the half-duplex mode.	The Full-duplex mode has better performance among simplex and half-duplex mode as it doubles the utilization of the capacity of the communication channel.
Example	Examples of Simplex mode are radio, keyboard, and monitor.	Example of half-duplex is Walkie-Talkies.	Example of the Full-duplex mode is a telephone network.

Transmission media

- Transmission media is a communication channel that carries information from sender to receiver. Data is transmitted through electromagnetic signals.
- The main functionality of the transmission media is to carry information in form of bits through **LAN**(Local Area Network).
- It is a physical path between transmitter and receiver in data communication.
- The electrical signals can be sent through copper wire, fibre optics, atmosphere, water, vacuum.
- Transmission media is of two types - wired media and wireless media.
- Different transmission media have different properties such as bandwidth, delay, cost and ease of installation and maintenance.

Classification Of Transmission Media:



Guided Media

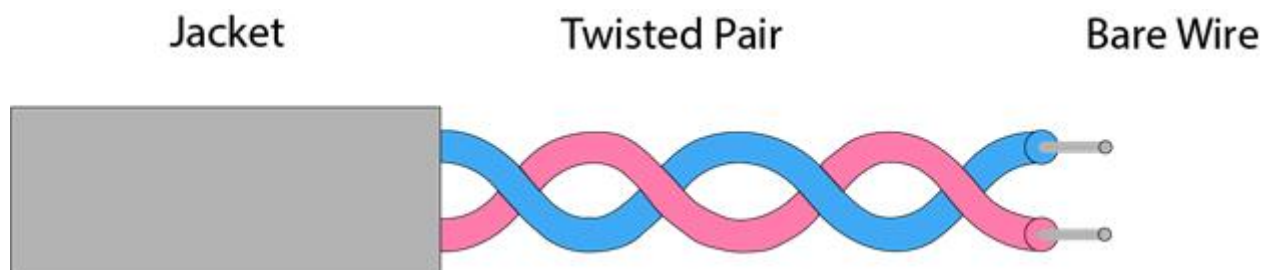
It is defined as the physical medium through which the signals are transmitted. Its also called Bounded media.

Types Of Guided media

1. Twisted pair

Twisted pair is a physical media made up of a pair of cables twisted with each other. A twisted pair cable is cheap as compared to other transmission media. Installation of the twisted pair cable is easy, and it is a lightweight cable.

A twisted pair consists of two insulated copper wires arranged in a regular spiral pattern.



Types of Twisted pair

a) Unshielded Twisted Pair

An unshielded twisted pair is widely used in telecommunication. The categories of the unshielded twisted pair cable:

- **Category 1:** Category 1 is used for telephone lines that have low-speed data.

Advantages Of Unshielded Twisted Pair:

- It is cheap.
- Installation of the unshielded twisted pair is easy.
- It can be used for high-speed LAN.

Disadvantage:

- This cable can only be used for shorter distances because of attenuation.

b)Shielded Twisted Pair

A shielded twisted pair is a cable that contains the mesh surrounding the wire that allows the higher transmission rate.

Advantages Of Shielded Twisted Pair:

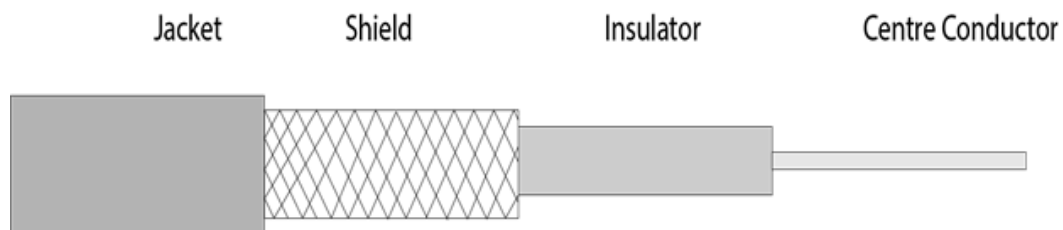
- An installation of STP is easy.
- It has higher capacity as compared to unshielded twisted pair cable.
- It has a higher attenuation.
- It provides the higher data transmission rate.

Disadvantages

- It is more expensive as compared to UTP and coaxial cable.
- It has a higher attenuation rate.

2.Coaxial Cable

- Coaxial cable is very commonly used transmission media, for eg TV wire is usually a coaxial cable.
- The name of the cable is coaxial as it contains two conductors parallel to each other.
- The inner conductor of coaxial cable is made up of copper and outer conductor is made up of copper mesh. The middle core is made up of non-conductive cover that separates inner conductor from the outer conductor.
- The middle core is responsible for data transferring whereas copper mesh prevents from the **EMI**(Electromagnetic interference).



Coaxial cable is of two types:

1. **Baseband transmission:** The process of transmitting a single signal at high speed.
2. **Broadband transmission:** The process of transmitting multiple signals simultaneously.

Advantages Of Coaxial cable:

- The data can be transmitted at high speed.

- It has better shielding as compared to twisted pair cable.
- It provides higher bandwidth.

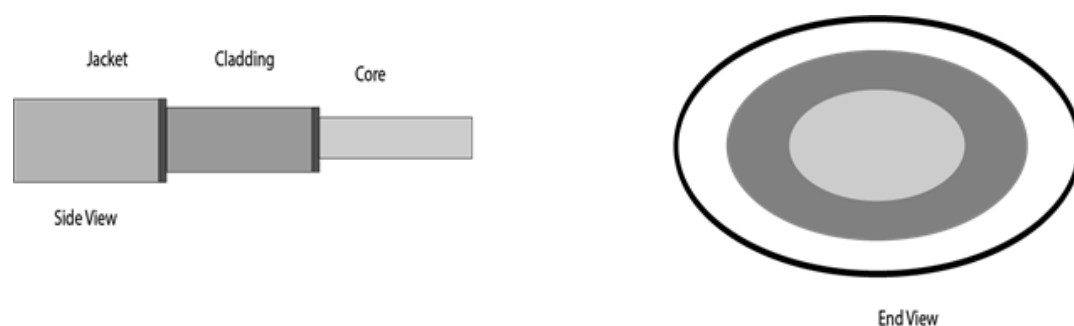
Disadvantages Of Coaxial cable:

- It is more expensive as compared to twisted pair cable.
- If any fault occurs in the cable causes the failure in the entire network.

3.Fibre Optic

- Fibre optic cable is a cable that uses electrical signals for communication.
- Fibre optic is a cable that holds the optical fibres coated in plastic that are used to send the data by pulses of light.
- The plastic coating protects optical fibres from heat, cold, electromagnetic interference.
- Fibre optics provide faster data transmission than copper wires.

Diagrammatic representation of fibre optic cable:



Basic elements of Fibre optic cable:

- **Core:** The optical fibre consists of a narrow strand of glass or plastic known as a core. A core is a light transmission area of the fibre. The more the area of the core, the more light will be transmitted into the fibre.
- **Cladding:** The concentric layer of glass is known as cladding. The main functionality of the cladding is to provide the lower refractive index at the core interface as to cause the reflection within core so that light waves are transmitted through the fibre.
- **Jacket:** The protective coating consisting of plastic is known as a jacket. The main purpose of a jacket is to preserve fibre strength and fibre protection.

Advantages of fibre optic cable over copper:

- **Greater Bandwidth:** The fibre optic carries more data as compared to copper cable.
- **Faster speed:** Fibre optic cable carries the data in the form of light. This allows the fibre optic cable to carry the signals at a higher speed.
- **Longer distances:** The fibre optic cable carries data at a longer distance

- **Better reliability:** The fibre optic cable is more reliable as it is immune to any temperature changes.
- **Thinner and lighter:** Fibre optic cable is thinner and lighter in weight so it can bear more pull pressure than copper cable.

UnGuided Transmission

- Unguided transmission transmits electromagnetic waves without using any physical medium. Therefore it is also called **wireless transmission**.
- In unguided media, air is the media through which the electromagnetic energy can flow easily.

Classification of Unguided transmission

1. Radio waves

- Radio waves are the electromagnetic waves that are transmitted in all the directions of free space.
- Radio waves are omnidirectional, i.e., the signals are propagated in all the directions.
- The range in frequencies of radio waves is from 3Khz to 1 khz.
- In the case of radio waves, the sending and receiving antenna are not aligned, i.e., the wave

sent by the sending antenna can be received by any receiving antenna.

- An example of the radio wave is **FM radio**.

Applications Of Radio waves:

- A Radio wave is useful for multicasting when there is one sender and many receivers.
- An FM radio, television, cordless phones are examples of a radio wave.

Advantages Of Radio transmission:

- Radio transmission is mainly used for wide area networks and mobile cellular phones.
- Radio waves cover a large area, and they can penetrate the walls.
- Radio transmission provides a higher transmission rate.

2.Microwaves

- Microwave transmission is a technology that transmits the focused beam of a radio signal from one ground-based microwave transmission antenna to another.
- Microwaves are the electromagnetic waves having frequency in the range from 1GHz to 1000 GHz.

- Microwaves are unidirectional as the sending and receiving antenna is to be aligned, i.e., the waves sent by sending antenna are narrowly focussed.
- In this , antennas are mounted on towers to send a beam to another antenna which is km away.
- It works on line of sight transmission, i.e antennas mounted on towers are direct sight of each other.

Advantages Of Microwave:

- Microwave transmission is cheaper than using cables.
- It is free from land acquisition as it does not require any land for the installation of cables.
- Microwave transmission provides an easy communication in terrains as the installation of cable in terrain is quite a difficult task.
- Communication over oceans can be achieved by using microwave transmission.

Disadvantages of Microwave transmission:

- **Eavesdropping:** An eavesdropping creates insecure communication. Any malicious user can catch the signal in the air by using its own antenna.
- **Out of phase signal:** A signal can be moved out of phase by using microwave transmission.

- **Susceptible to weather condition:** This means that any environmental change such as rain, wind can distort the signal.

3.Infrared

- An infrared transmission is a wireless technology used for communication over short ranges.
- The frequency of the infrared is in the range from 300 GHz to 400 THz.
- It is used for short-range communication such as data transfer between two cell phones, TV remote operation, data transfer between a computer and cell phone resides in the same closed area.

Characteristics Of Infrared:

- It supports high bandwidth, and hence the data rate will be very high.
- Infrared waves cannot penetrate the walls. Therefore, the infrared communication in one room cannot be interrupted by the nearby rooms.
- An infrared communication provides better security with minimum interference.
- Infrared communication is unreliable outside the building because the sun rays will interfere with the infrared waves.

Internet

- Internet is a global network that connects billions of computers across the world with each other and to the World Wide Web.
- It uses standard internet protocol suite (TCP/IP) to connect billions of computer users worldwide.
- It is set up by using cables such as optical fibers and other wireless and networking technologies.
- Internet is the fastest mean of sending or exchanging information and data between computers across the world.
- Internet is called a network as it creates a network by connecting computers and servers across the world using routers, switches and telephone lines, and other communication devices and channels.
- It can be considered a global network of physical cables such as copper telephone wires, fiber optic cables, tv cables, etc.
- Furthermore, even wireless connections like 3G, 4G, or Wi-Fi make use of these cables to access the Internet.

Applications of Internet

1. Communication
2. Web Browsing
3. Online Shopping
4. Real-Time Update
5. Social Media
6. Job Search
7. Education
8. Travel
9. Stock Market Update
10. Video Conferencing

1. Communication

Communication refers to exchanging ideas and thoughts between people to create understanding. In organizations, both formal and informal communications take place. Formal communications refer to official communications in orders, notes, circulars, agenda etc. Informal communications are usually in form of rumors, whispers, etc.

2. Web Browsing

Web Browsing is one of the applications of the internet. A web browser is a program that helps the user to interact with all the data in the WWW (World

Wide Web). There are many web browsers present in today's world. Some of them are Google Chrome Safari

3. Online Shopping

The shops provide all the necessary details of a product on their website, so the user can choose as per their needs.

4. Real-Time Update

The internet makes things easier. One can quickly get an update on things happening in real-time in any part of world. For example sports, politics, business.

5. Social Media

Social media is a place where user can communicate with anyone, like friends, family, classmates, etc. User can promote their businesses on social media as well.

6. Job Search

The candidate can search for their dream job, apply and get it very easily. Even companies nowadays post their need on the internet and hire candidates as per their skills based on the job role like LinkedIn.

7. Education

The Internet has a vital role in the education field. It became an effective tool in both teaching and learning. Teachers can upload their notes or learning videos on the websites with the help of the internet.

8. Travel

Users can easily search for their favourite tourist places worldwide and plan their trips. One can book holiday trips, cabs, hotels, flight tickets, clubs, etc., with the help of the Internet. Some websites that provide these facilities are goibibo.com , makemytrip.com

9. Stock Market Update

The stock market is where individuals buy and sell publicly traded company shares. Stock market updates include vital data and statistics, like the current prices of major stocks, individual stock prices, trading volumes, market capitalization, and price movements.

10. Video Conferencing

It allows users in different locations to hold face-to-face meetings. Video conferencing is a widely accepted mode of communication among businesses, houses, and other organizations.

Hardware Requirements of the Internet

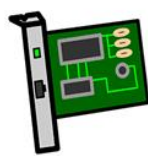
Any one of the following must be present in order to connect to the Internet.

- The Internet is connected over a telephone connection using a **Modem**. A modem is the main piece of gear you require. The modem you require depends on the type of Internet access you select.

- The most crucial piece of hardware needed to connect to the Internet is the **NIC - Network Interface Card** (wired/wireless) facility. For instance, the laptop might be tethered or wirelessly connected to the internet.
- Utilising a cellular network, a **dongle** is used to connect to the Internet.
- A **Wi-Fi router or hotspot** is used when using a wireless network to access the Internet. A router is a piece of hardware that enables you to create a home network or many computers and other devices connected to a single Internet connection. You may build a home wireless network, sometimes known as a Wi-Fi network, because many routers are wireless.
- Cellular **network-capable electronic device**.
- Internet connectivity through wired and wireless networks, cable TV, ISDN and dial-up connections.



Modem



NIC



Repeater



Hub



Switch



Router



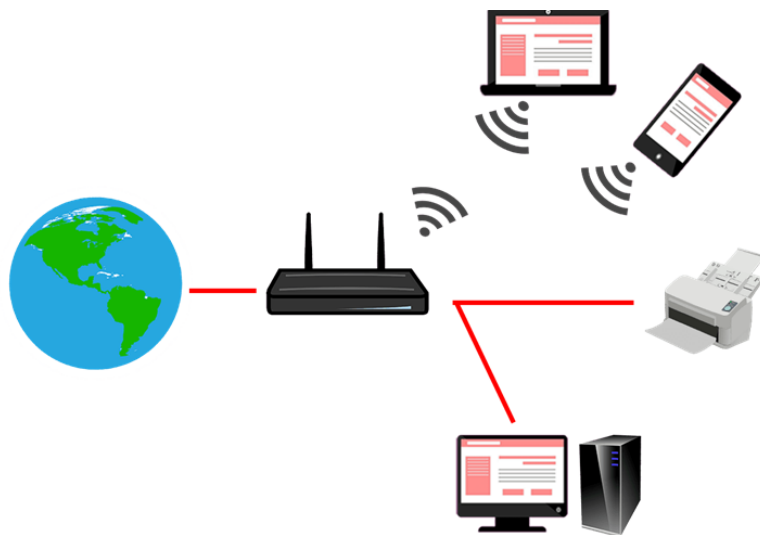
Bridge



Gateway

Software Requirement

- The operating system should support the **TCP/IP (Transfer Control Protocol)** and **HTTP (HyperText Transfer Protocol Secure) protocols**, as well as SMTP (Simple Mail Transfer Protocol), FTP (File Transfer Protocol), and HTTP (HyperText Transfer Protocol).
- Access to web apps like Outlook, Gmail, Whatsapp, Facebook, Twitter, and others through browsers and other Internet clients.



Other Requirements for Internet Connection Setup

You require a computer, a connecting device, and an ISP in order to access the internet (ISP). An ISP is a firm that connects people, companies, and organisations to the internet. Additionally, it could offer services like storage space for your personal data.

You can physically connect to your network connection device using a cable or a wireless method. This wireless method will often use the Wi-Fi protocol. The Wi-Fi standard is compatible with the majority of wireless devices.

Intranet

- An intranet is a private network to access information.
- An intranet belongs to an organization which is only accessible by **organization's employee**
- Main aim of intranet is to share the information and resources among the organization employees.
- An intranet provides the facility to work in groups and for teleconferences.
- A modern intranet is cloud-based, user-friendly, and easily accessible by all employees, including remote workers.

Applications of an Intranet

1. Improves internal communication

A modern intranet aims to encourage communication across your organization

- Teams can have open discussions

- Individuals can share knowledge
- Leadership can stay in touch
- Departments can provide updates

And the best part is that communication flows both ways. Anyone in the company can share information and articles. everyone can provide feedback.

2. Connects company across locations and time zones

It's a place where conversations occur, company news is shared, and events are planned. Each employee and location can see what's happening throughout the company, regardless of where they're based.

3. Helps employees find information

Intranets can become source of information for a company or business, where their employees can find every updates.

4. Boosts recognition and reward

Recognition is an essential part of the employee experience, and it serves to improve employees' satisfaction and engagement levels. Intranets could make this easier if it has a live feed or an integration with a peer-to-peer recognition tool like Bonusly.

5. Simplifies employee onboarding

New hires can use it to find their feet with easy-to-access documents, up-to-date organization charts, rich

bios on their peers, and a news hub that provides information and insights into the company's culture.

6. Provides organizational clarity

Company intranets with clear and dynamic organization charts provide clarity on set-up of your company. This clarity help employees understand the roles and responsibilities of individuals throughout your business.

7. Bring your people together

Receive a weekly email with our latest guides on navigating the modern workplace.

8. Encourages knowledge sharing

Intranets aim to allow organic knowledge sharing to grow as the person-to-person connections increase and deepen across your company. In this way, the company's knowledge can grow rapidly even with remote worker and more people share their learnings.

8. Reinforces your brand and values

- Publicly recognizing employees for living out your unique company values
- Sharing web articles so people can celebrate milestones
- Asking employees to share examples of the values they see in daily work life

9. Reduces emails and meetings

Intranets with a good, integrated instant chat feature make discussions easy. It improves communication, it could make it easier to keep everyone informed as well as garner feedback.

10. Brings your workplace culture to life

Intranets are supposed to bring your people together, enriching your company culture in the process.

11. Improves employee engagement

The goal of engaging employees is to make them happier, enjoy improved productivity, and be less likely to leave. An intranet could improve engagement in a number of ways.

Intranet advantages:

- **Communication:** It provides a cheap and easy communication. An employee of organization can communicate with other employee through email.
- **Time-saving:** Information on the intranet is shared in real time, so it is time-saving.
- **Collaboration:** Collaboration is one of the most important advantage of intranet. The information is distributed among employees of organization and can only be accessed by the authorized user.

- **Platform independency:** It is a neutral architecture as the computer can be connected to another device with different architecture.
- **Cost effective:** People can see data and documents by using the browser and distributes the duplicate copies over the intranet. This leads to a reduction in the cost.

S.NO	Internet	Intranet
1.	Internet is used to connect different networks of computers simultaneously.	Intranet is owned by private firms.
2.	On the internet, there are multiple users.	On an intranet, there are limited users.
3.	Internet is unsafe.	Intranet is safe.
4.	On the internet, there is more number of visitors.	In the intranet, there is less number of visitors.
5.	Internet is a public network.	Intranet is a private network.
6.	The Internet provides unlimited information.	Intranet provides limited information.
7.	It is used for communication, sharing of information, e-commerce, education, entertainment, and other purposes.	An intranet is primarily used for internal communication, collaboration, and information sharing within an organization.
8.	Examples of Internet-based services include email, social media, search engines, and online shopping sites.	Examples of intranet-based services include internal communications, knowledge management systems, and collaboration tools

